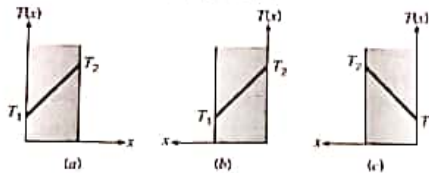


Problem Set #3 Sedra Khan.

Monday, January 27, 2020 4:24 PM

Do problems 2.7, 2.19, 2.20, and 2.40 from our textbook (Bergman et al.; 2019), on pages P-14 - P-19.

2.7 Consider a plane wall 120 mm thick and of thermal conductivity 120 W/m · K. Steady-state conditions are known to exist with $T_1 = 500$ K and $T_2 = 700$ K. Determine the heat flux q'' and the temperature gradient dT/dx for the coordinate systems shown.



$$q'' = \frac{q}{A} = -k \frac{dT}{dx} \quad \text{heat flux}$$

(a) $-120 \frac{\text{W}}{\text{m} \cdot \text{K}} \cdot \frac{(700 - 500)}{0.120 \text{ m}} = -200,000 \frac{\text{W}}{\text{m}^2}$
temp. grad: $-200,000 / 120 \text{ m} = 1666.7 \text{ W/m}^2$

(b) $-120 \frac{\text{W}}{\text{m} \cdot \text{K}} \cdot \frac{(500 - 700)}{0.120 \text{ m}} = +200,000 \frac{\text{W}}{\text{m}^2}$
temp. grad: $200,000 / 120 \text{ m} = -1666.7 \text{ W/m}^2$

(c) $-120 \frac{\text{W}}{\text{m} \cdot \text{K}} \cdot \frac{(700 - 500)}{0.120 \text{ m}} = -200,000 \frac{\text{W}}{\text{m}^2}$
temp. grad: $-200,000 / 120 \text{ m} = 1666.7 \text{ W/m}^2$

2.19 Consider a one-dimensional plane wall of thickness $2L$ that experiences uniform volumetric heat generation. The surface temperatures of the wall are maintained at $T_{s,1}$ and $T_{s,2}$ as shown in the sketch. Verify, by direct substitution, that an expression of the form

$$T(x) = \frac{\dot{q} L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \frac{T_{s,2} - T_{s,1}}{2} \frac{x}{L} + \frac{T_{s,1} + T_{s,2}}{2}$$

satisfies the steady-state form of the heat diffusion equation. Determine an expression for the heat flux distribution, $q''(x)$.



Evaluate @ B.C ① $T = T_{s,1}$ and $x = -L$

$$T_1 = -\dot{q}(-L)^2 \cdot \frac{1}{2k} + c_1(-L) + c_2$$

② $T = T_{s,2}$ and $x = L$

$$T_2 = -\dot{q}(L)^2 \cdot \frac{1}{2k} + c_1(L) + c_2$$

Starting with the general form of the heat diffusion equation:

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c \frac{\partial T}{\partial t}$$

$$\frac{\partial^2 T}{\partial x^2} (k) + \dot{q} = 0 \rightarrow \int \frac{\partial^2 T}{\partial x^2} \cdot k = \int -\dot{q}$$

$$= \frac{dT}{dx} = \frac{-\dot{q}}{k} x + c_1$$

integrate, again... $T = -\dot{q} x^2 \cdot \frac{1}{2k} + c_1 x + c_2$

$$T_1 = -\dot{q}(-L)^2 \cdot \frac{1}{2k} - c_1 L + T_2 + \dot{q}(L)^2 \cdot \frac{1}{2k} - c_1 L$$

$$\therefore \boxed{c_1 = \frac{T_1 - T_2}{-2L}} \quad \therefore \frac{2T_2}{2} + \frac{\dot{q}}{2} \frac{L^2}{k} + \left(\frac{T_1 - T_2}{-2L} \right)$$

using our constants, we found:

$$\frac{T_1 + T_2}{2} + \frac{\dot{q}}{2} \frac{L^2}{k} = c$$

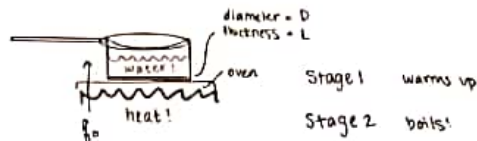
$$T(x) = \frac{\dot{q} L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \left(\frac{T_2 - T_1}{2} \right) \frac{x}{L} + \left(\frac{T_1 + T_2}{2} \right)$$

the heat flux q'' is simply $\frac{dT}{dx} \cdot (-k)$

$$\hookrightarrow \frac{dT}{dx} \text{ shown above} = -\frac{\dot{q}}{k} x + c_1 = -\frac{\dot{q}}{k} x + \frac{T_1 - T_2}{-2L}$$

$$\hookrightarrow -k \cdot \frac{dT}{dx} = \dot{q} x + \frac{(T_1 - T_2)k}{-2L} = q''$$

2.20 A pan is used to boil water by placing it on a stove, from which heat is transferred at a fixed rate q_0 . There are two stages to the process. In Stage 1, the water is taken from its initial (room) temperature T_i to the boiling point, as heat is transferred from the pan by natural convection. During this stage, a constant value of the convection coefficient h may be assumed, while the bulk temperature of the water increases with time, $T_w = T_w(t)$. In Stage 2, the water has come to a boil, and its temperature remains at a fixed value, $T_w = T_w$, as heating continues. Consider a pan bottom of thickness L and diameter D , with a coordinate system corresponding to $x = 0$ and $x = L$ for the surfaces in contact with the stove and water, respectively.



Stage 1 warms up

Stage 2 boils

STAGE 1

thermal conductivity is constant

$$\text{heat equat } \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \text{ where } \alpha = \frac{k}{\rho c_p}$$

initial conditions:
 $T(x, 0) = T_i$

b.c Finite Heat Flux $q_0 = -k \frac{\partial T}{\partial x} \big|_{x=0} = q_0 / (\pi D^2 / 4)$

Convection Surface Conditions $-k \frac{\partial T}{\partial x} \big|_{x=L} = h [T_L - T_w]$

STAGE 2

$$\text{heat equat } \frac{\partial^2 T}{\partial x^2} = 0$$

b.c $-k \frac{\partial T}{\partial x} \big|_{x=0} = \frac{4q_0}{\pi D^2}, T(L) = T_w$

2.40 A spherical shell of inner and outer radii r_i and r_o respectively, contains heat-dissipating components, and at a particular instant the temperature distribution in the shell is known to be of the form

$$T(r) = \frac{C_1}{r} + C_2$$

Are conditions steady-state or transient? How do the heat flux and heat rate vary with radius?

Spherical coordinates:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(K r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} \left(K \sin^2 \theta \frac{\partial T}{\partial \phi} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(K \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{\partial T}{\partial t} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\therefore \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \left[\frac{C_1}{r} + C_2 \right] \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$-\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{C_1}{r^2} \right) = 0 \therefore \text{STEADY STATE.}$$

heat rate $q = -KA \frac{\partial T}{\partial r} = -K(\pi r^2) \cdot \frac{C_1}{r^2} = C_1 \cdot \pi \cdot K$ it does not vary w/ radius!

heat flux $q = -k \frac{\partial T}{\partial r} = -\frac{K C_1}{r^2}$ as radius increases, flux decreases